

Continuity, Convexity, Centrality: Axiomatizing and Comparing Location Indices^{*}

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Abstract

I provide an axiomatic foundation of location indices: single values that summarize the position of finite distributions (datasets). I characterize the class of location indices that are minimizers of a continuous and convex expected loss. The main result establishes six axioms that are necessary and sufficient for such a representation, with the corresponding loss unique up to a positive multiplicative constant. The recovered class includes a broad set of summary statistics, such as the mean and — in the limit — the median, but not the mode. Since the loss is recovered from observed behavior, it can be interpreted through a decision-theoretic lens: I show that, under suitable assumptions, the degree of asymmetry at the origin of the loss function can be seen as a measure of first-order risk attitudes, while the loss function's degree of convexity governs the sensitivity of the index to extreme observations. I also show that, for any dataset, any returned index coincides with one of its expectiles; so, a robust index is observationally indistinguishable from a sensitive one with a biased asymmetry level.

Keywords— Asymmetry, average, estimation, expectiles, quantiles, robustness

JEL codes— C91, D81, D83, D87

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I. INTRODUCTION

Replacing a whole distribution of numbers with a representative single value is one of the most common operations in Economics: we talk about average income, median voters, point forecasts, certainty equivalents, and so on. The choice of a summary statistic rather than others is usually settled on pragmatic grounds: the mean is intuitive, the median is robust, the mode is easy. This choice is not innocuous: reporting the mean of a right-skewed income distribution rather than its median is a decision that lets a few rich individuals pull the reported measure up, and treats upward approximations as no worse than downward ones of the same size. More generally, when choosing any summary statistic, one implicitly takes a stand on how mistakes of different signs and magnitudes should be penalized.

In this paper, I study *location indices*: maps m that assign to each dataset $D = \{\mathbf{x}; \mathbf{p}\}$ a single value $m(D)$ describing its central position. Many common indices are found as minimizers of an expected loss:

$$m(D) = \arg \min_y \sum_i p_i c(x_i, y),$$

where $c(x, y)$ is a continuous and convex function measuring the cost of representing an observation x with the value y . This is not only true for traditional statistical methods — e.g. the least squares criterion, or the maximum log-likelihood principle — but also for more recent algorithms from the machine learning literature: the Hedge algorithm, the Online Gradient Descent (OGD), Follow the Regularized Leader (FTRL), Support Vector Machines (SVM) all usually assume a continuous, convex loss function.¹ Read this way, c is not a mere technical device; rather, it serves as a description of the analyst’s preferences over mistakes: how much a given over- or under-approximation hurts.

While the elicibility literature (Osband, 1985) relates functionals to losses by taking the functional as given and asking which losses are consistent with it, I reverse the direction by placing interpretable axioms on the map itself, investigating which of them admit a representation in terms of an expected loss. To some extent, this can be seen as a revealed preference exercise over averages: what must an averaging choice m

¹Indeed, it is common in supervised learning to use continuous, convex surrogates to replace the discontinuous or non-convex ideal objective.

satisfy for it to be as if minimizing an expected loss for some well-behaved c ?

The answer to this question is Theorem 1, in which I pin down six axioms — Certainty Matching, Continuity, Dominance, Translation Invariance, Odds Independence, and Reduction — that are jointly necessary and sufficient for m to minimize a continuous, strictly convex, translation-invariant expected loss, and I show that the corresponding loss function is unique up to a positive multiplicative constant. The real novelty of such representation consists in the last two axioms, as the first four are regularity conditions shared, in one guise or another, with classical theories of means. In particular, while Odds Independence disciplines how observations and relative frequencies are traded off in binary datasets — allowing to rationalize the map with a single loss rather than with a reference-dependent family — Reduction imposes a recursive structure to the evaluation of datasets, guaranteeing that the representation built on binary ones can be extended to those of arbitrary size.

The class of summary statistics recovered by these axioms is not new, as it coincides with the translation-invariant M -estimators of Huber (1964). What is new is the direction of the argument, together with its implied behavioral content: since the loss is recovered from the map rather than assumed, its geometrical features become properties inherited from the valuation choices rather than modeling assumptions. In particular, I focus on how the loss weighs over- against under-prediction, which corresponds to a notion of *asymmetry* of the loss function with respect to the vertical axis, and how steeply it grows with the size of a mistake, which corresponds to its degree of *convexity*. I show that the asymmetry at the origin governs first-order risk attitudes in the sense of Segal and Spivak (1997): a loss that punishes small downside errors more sharply than equal upside ones produces a conservative measure of centrality; hence, *risk aversion*. Similarly, I tie the notion of convexity with the one of robust statistics by relating the degree of curvature of the loss function with the gross error sensitivity of m .

While conceptually distinct, asymmetry and convexity may be difficult to disentangle in the data: in Theorem 2, I show that for any finite dataset and reported summary statistic, there exists an asymmetry level so that the chosen value corresponds to an expectile of the dataset; as a result, a robust index evaluated on one dataset is observationally indistinguishable from a sensitive one applied with a biased directional weight. This means that sensitivity to extremes observations cannot be read off a single evaluation, and that

asymmetry and convexity cannot be jointly identified from any cross-sectional forecast data.

The remainder of this paper proceeds as follows. Section 2 reviews the related literature. Section 3 sets up the model. Section 4 introduces the axioms, proves the representation theorem, and locates the mean, median, and mode within the framework. Section 5 develops the decision-theoretic analysis of asymmetry and convexity. Section 6 concludes.

II. RELATED LITERATURE

This paper sits at the intersection between the mathematical theory on means and representative values, the statistical literature on elicibility and consistent scoring functions, and the decision-theoretic characterization of certainty equivalents. In all such studies, the main object of interest is a location measure representable as the minimizer of an expected loss.

The formal study of representative values originates with Cauchy (1821), who introduced the concept of average (or mean) as any intermediate value that lies strictly between the minimum and maximum values of a given set. To some extent, this is the first example of axiomatic characterization of a location index, as Cauchy defined the notion of mean by its desirable properties rather than simply providing a mathematical formula. Since then, several axiomatic works on different notions of means have been developed (Kolmogorov, 1930; Nagumo, 1930; Chambers, 2009; Bellini and Peri, 2022). For instance, Kolmogorov (1930) and Nagumo (1930) independently characterized quasi-linear means through the properties of reflexivity, continuity, monotonicity, symmetry, and associativity.

A parallel stream of work, associated with Chisini (1929) and De Finetti (1931), defined a mean through an invariance property: a value that, when substituted for every observation, leaves a designated functional of the data unchanged. Such property is variational in the spirit, and is the direct ancestor of the Reduction axiom proposed in this study. Larger discussions on the concept and multiplicity of means are discussed by De Carvalho (2016) and Singpurwalla and Lai (2020).

Theorem 1 in this paper extends this work in two ways. First, it enlarges the admissible class beyond the quasi-linear means to the full set of indices elicitable by a continuous, strictly convex loss. Second, and more crucially, it abandons the symmetry axiom: by permitting the loss to weight positive and negative deviations

differently, the characterization admits asymmetric indices — such as expectiles, and quantiles in the limit — that the classical theory excludes by construction. The Reduction axiom plays, in this setting, the role occupied by associativity in the quasi-linear theory, and provides a more flexible notion of invariance.

In the language of forecast evaluation, the representation in this paper, $m(D) = \arg \min_y \sum_i p_i c(x_i, y)$, means that the index m is *elicitable*, and that c is a strictly consistent scoring function for it. The question of which functionals admit such a representation — and which scores are consistent for a given functional — has been studied extensively in statistics and, more recently, in mathematical finance: Savage (1971) and Schervish (1989) characterized the scores consistent for the mean, Osband (1985) established the principle linking consistent scores to identification functions; and Gneiting (2011) unified these results, showing that the mean is elicited precisely by Bregman divergences, quantiles by generalized piecewise-linear scores, and expectiles by their asymmetric analogues. Lambert et al. (2008) and Steinwart et al. (2014) characterized elicibility itself through the convexity of the functional’s level sets, while Ziegel (2016) and Fissler and Ziegel (2016) extended the analysis to coherent risk measures and to higher-order elicibility. Within Economics, Elliott et al. (2005) developed the estimation of forecasts under flexible loss.

This paper contributes to this literature by proposing a dual, complementary approach to the topic: instead of taking the functional as primitive and asking which scores are consistent for it, my characterization proceeds in reverse, placing interpretable axioms on the location map, and recovering the class of functionals it can be represented by. While the recovered class is not itself new — it corresponds to the translation-invariant location M -estimators of Huber (1964) — the axiomatic foundation is, and the substantive content concentrates in the two axioms without a counterpart in classical theories, Odds Independence and Reduction.

This paper is also close to some work in decision theory interpreting certainty equivalents as a measure of centrality for probability distributions of prizes. While the intuition is well-grounded (Huygens, 1657; Bernoulli, 1738; Neumann and Morgenstern, 1944; Finetti and Emanuelli, 1967), the strategy of resolving to this parallelism to shed light on decision under risk is relatively new. For instance, Hong (1983) moved beyond quasi-linearity to accommodate income inequality and the Allais paradox, while Fishburn (1986) provided an axiomatic characterization of certainty equivalents as implicit mean values, moving beyond expected utility representations. This paper is methodologically closest to the latter: Theorem 1 can be seen

as a modification of Fishburn’s representation in which skew-symmetry and ratio-continuity are relaxed and uniform monotonicity is forced to be positive, so that the resulting indices are genuine location measures rather than signed evaluations. This strand also licenses the decision-theoretic reading developed in Section 5, in which the loss function is the primitive expression of a decision maker’s preferences over mistakes.

Finally, the paper is also related to the literatures on specific measures of centrality. In particular, it connects to the concept of expectiles (Newey and Powell, 1987), which have been often referred to as generalized quantiles (Breckling and Chambers, 1988; Chen, 1996; Bellini, Klar, et al., 2014), and applied to risk management (Bellini and Di Bernardino, 2017; Daouia et al., 2020). Quantiles, characterized axiomatically by Chambers (2009) and Bellini and Peri (2022) and estimated by Koenker and Bassett Jr (1978), enter the present framework as a limiting case.

III. MODEL

Let $D = \{\mathbf{x}; \mathbf{p}\}$ denote a *dataset*, where $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ is a set of *observations*, and $\mathbf{p} = (p_1, \dots, p_n) \in (0, 1]^n$ is the set of corresponding *frequencies*. I assume without loss of generality that observations are increasingly ordered, i.e. $x_1 < \dots < x_n$, and that frequencies are expressed in relative terms, i.e. $\sum_{i=1}^n p_i = 1$. The space of all such datasets is $\mathcal{D}$, while the one of all D with $n \leq 2$ (i.e. degenerate and binary datasets) is $\tilde{\mathcal{D}} \subset \mathcal{D}$. Degenerate distributions $\{x; 1\}$ are denoted by δ_x .² Finally, throughout the analysis, let $D + k = \{x_1 + k, \dots, x_n + k; p_1, \dots, p_n\}$ for any $k \in \mathbb{R}$, and $\mathbb{N}_D = \{1, \dots, n\}$ denote the index set of dataset D .

A *location index* for D is a single value that describes the dataset’s central position. Common examples of location indices are the mean, the median, the mode, the minimum, the maximum, and so on. Formally, I model the determination of a location measure (informally called “averaging”) through a *location mapping* $m : \mathcal{D} \rightarrow \mathbb{R}$. Note how this process can be seen as the approximation of $D = \{x_1, \dots, x_n; p_1, \dots, p_n\}$ with $\delta_{m(D)} = \{m(D); 1\}$. The cost of this operation is represented by the consequential loss of information: when averaging data, I obtain a coarser yet easier description of its distribution.³ The cost of approximating

²I metrize $\tilde{\mathcal{D}}$ by identifying each binary dataset with the corresponding parameter vector $(x_1, x_2, p) \in \mathbb{R}^2 \times (0, 1)$, while each degenerate dataset δ_x is identified by $(x, \cdot, 1)$. The resulting set is endowed with the Euclidean distance, and convergence in Axiom 2 is convergence in this metric.

³This roughly corresponds, at least in the core intuition, to what occurs when converting an image from

observation x with value y is measured by a *loss function* $c(x, y)$.

Definition 1 $c : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}_0^+$ is a *loss function* if it is continuous in each argument, strictly convex in its second argument, $c(x, y) = 0 \Leftrightarrow x = y$, and $c(x, y) = c(x + k, y + k)$ for any $k \in \mathbb{R}$.

Continuity is a technical assumption. $c(x, y) = 0 \Leftrightarrow x = y$ simply states that there is no loss of information when (and only when) no approximations have been made. $c(x, y) = c(x + k, y + k)$ — which I will refer to as functional translation invariance — implies that the cost of the information loss caused by approximating x with y only depends on the size of the mistake $(x - y)$, while its position on the real line is irrelevant.⁴ Strict convexity implies that larger errors amplify the penalty disproportionately — each additional unit of deviation from x is more expensive than the previous one. Together with the normalization at $x = y$, it also means that bigger mistakes (in either directions) cost more. Finally, paired with functional translation invariance, strict convexity characterizes c as a measure of difference. The set of all such loss functions is denoted by $\mathcal{C}$.

For a given $c \in \mathcal{C}$, I define the cost of representing D with y as:

$$C(D, y) := \sum_{i=1}^n p_i c(x_i, y) \quad (1)$$

This is consistent with standard methodologies in statistics and ML, and enables the model to incorporate OLS and WLS as special cases.

IV. AXIOMATIZATION

I am interested in which properties the location mapping must satisfy so that the corresponding location index $m(D)$ is a minimizer of $C(D, y)$ for some $c \in \mathcal{C}$. In other words, I am looking for all measures of central tendency that can be characterized as the solution of a variational problem, i.e. that are minimizing variation from the center according to some notion of discrepancy.⁵

PNG to JPG format: the image quality is sacrificed in order to get a lighter file.

⁴While ensuring to capture the position of the distribution on the real line (and only that!), functional translation invariance rules out relative penalties, e.g. $c(x, y) \neq c(\lambda x, \lambda y)$ in general.

⁵Computing a location index $m(D) = y$ amounts to collapsing a non-degenerate distribution D onto the degenerate distribution δ_y . This operation transports mass p_i from each x_i to y at unit cost $c(x_i, y)$, so that

To start, I introduce an axiom on degenerate datasets:

Axiom 1 (Certainty Matching) $m(\delta_x) = x$

Certainty matching states that the location index of a point observation is the observation itself. This intuitive condition is satisfied by all known measures of central tendency.

Axiom 2 (Continuity) For any $\{D_k\}_{k \in \mathbb{N}} \subseteq \ddot{D}$ and $D \in \ddot{D}$, $D_k \rightarrow D \Rightarrow m(D_k) \rightarrow m(D)$

Continuity is a technical assumption which, broadly speaking, requires similar datasets to have similar measures of central tendency.^{6 7}

Axiom 3 (Dominance) $\forall D, D' \in \ddot{D}$, $D \succ_{FOSD} D' \rightarrow m(D) > m(D')$

Dominance requires binary distributions of unambiguously higher observations to have higher measures of centrality. To some extent, it complements the previous axioms by characterizing “reasonable” summary statistics: as matter of fact, Certainty matching, Continuity and Dominance are all included in Kolmogorov and Nagumo’s axioms on regular means (Kolmogorov, 1930; Nagumo, 1930).

Axiom 4 (Translation Invariance) $\forall k \in \mathbb{R}$ and $D \in \ddot{D}$, $m(D + k) = m(D) + k$

In words, adding k to every observation of a binary dataset translates it horizontally on the number line by k . As a result, the position of the dataset, measured by m , is also shifted by k , while relative positions within the dataset remain unchanged. Intuitively, this axiom ensures that the location index is a pure measure of central location tracking where the data is, rather than how it is shaped.

Axiom 5 (Odds Independence) Let $D = \{\underline{x}, \bar{x}, p\}$, $D' = \{\underline{x}, \bar{x}, q\}$, $D'' = \{\underline{x}, \bar{x}, r\}$ and $D''' = \{\underline{x}, \bar{x}, s\}$, with $\underline{x} < \underline{x} < y < \bar{x} < \bar{x}$. If $m(D) = m(D') = m(D'') = m(D''') = y$, then:

$$\frac{q}{1-q} \cdot \frac{1-p}{p} = \frac{s}{1-s} \cdot \frac{1-r}{r}$$

minimizing the total cost $C(D, y)$ is formally a one-dimensional optimal transport problem (Monge, 1781; Kantorovich, 1942). The optimal location index $m(D)$ minimizes this transport cost. The connection is not merely algebraic: psychological evidence suggests that humans represent numbers on a spatially oriented mental number line (Moyer and Landauer, 1967; Dehaene et al., 1999), providing substantial support to the “cost of moving mass” interpretation.

⁶Note that I require continuity on degenerate and binary datasets only.

⁷Axiom 2 is stated sequentially, whereas the Euclidean metric makes it an assertion of joint continuity of m in (x_1, x_2, p) . The two coincide: $\ddot{D}$ is a metric space, hence first-countable, so sequential continuity is equivalent to continuity.

Odds Independence regulates how observations and frequencies are traded off in binary datasets. In particular, the change in frequencies required to keep the average constant when increasing the higher observation should not depend on the lower observation, and viceversa. In other words, the “exchange rate” between value changes and frequency changes is independent across observations.

Axiom 6 (Reduction) For any D of size $n > 2$ with $m(D) = y$, there exists a dataset D' of size $(n - 1)$ with $m(D') = y$ obtained from D by one of the following two constructions.

1. **Reduction by Cancellation:** If $y = x_k$ for some $x_k \in \text{supp}(D)$, then:

$$D' = \left\{ \{x_i\}_{i \neq k}; \left\{ \frac{p_i}{1 - p_k} \right\}_{i \neq k} \right\}$$

2. **Reduction by Aggregation:** If $y \in (x_1, x_n)$ but $y \notin \text{supp}(D)$, D' is obtained by replacing an adjacent pair (x_i, x_{i+1}) lying on the same side of y with a single outcome $\tilde{x}$ of frequency $p_i + p_{i+1}$. To determine $\tilde{x}$, fix any anchor k strictly on the opposite side of y from the pair. Then, for each x on the pair’s side, let $s_y(x; k) \in (0, 1)$ be the (unique) weight on x such that $m\{s_y(x; k) \delta_x + [1 - s_y(x; k)] \delta_k\} = y$. Then, $\tilde{x}$ is the unique value satisfying:

$$\frac{1 - s_y(\tilde{x}; k)}{s_y(\tilde{x}; k)} = \frac{p_i}{p_i + p_{i+1}} \cdot \frac{1 - s_y(x_i; k)}{s_y(x_i; k)} + \frac{p_{i+1}}{p_i + p_{i+1}} \cdot \frac{1 - s_y(x_{i+1}; k)}{s_y(x_{i+1}; k)} \quad (2)$$

Reduction is a procedural axiom describing a recursive structure for simplifying datasets. This allows to iteratively reduce the size of a dataset while keeping its centrality fixed, in the same spirit of the associativity property of quasi-linear means (Kolmogorov, 1930; Nagumo, 1930). The actual simplification method depends on how the selected location index compares to the observations: if y is equal to one of the observations x_k , with $k \in \mathbb{N}_D$, then the axiom acts as a cancellation principle: D' is obtained by removing x_k from the dataset, and redistributing its mass p_k among the remaining $n - 1$ values in proportion to their original frequencies.⁸

⁸For instance, if $D = \{1, 2, 3; \frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}$ and $m(D) = 2$, then $D' = \{1, 3; \frac{1}{2}, \frac{1}{2}\}$.

Otherwise, a pair of adjacent observations is merged according to a similarity-preserving transformation.⁹

To see why, note that the weight $s_y(x; k)$ captures how similar the observation x is to the target measure y : the more weight x must carry to drag a two-point location index onto y (against the fixed opponent k), the closer x and y already are. Hence, the corresponding odds $\frac{1-s_y(x; k)}{s_y(x; k)}$ behave as a dissimilarity index, possibly on a non-linear scale, which is small for x close to y , and grows as x and y move apart.¹⁰

It is important to clarify the role of the anchor k : in this context, k fixes the unit in which dissimilarities are expressed, but not their relative magnitudes. Formally, Axiom 5 implies that moving from one anchor to another corresponds to multiplying all odds by the same conversion rate. Hence, Equation (2) remains the same regardless of k , and so does $\tilde{x}$.

The next result characterizes the class of location mappings satisfying Axioms 1–6:

Theorem 1 (Variational Representation) *The following are equivalent:*

1. m satisfies Axioms 1–6;
2. $\exists c \in \mathcal{C}$ such that:

$$m(D) = \arg \min_y \sum_{i=1}^n p_i c(x_i, y) \quad \forall D \in \mathcal{D}$$

Also, c is unique up to any positive multiplicative constant.

I call this *variational representation* because the index m is not defined by an explicit formula; rather, it is the solution of an optimization problem: $m(D)$ is the value from which the data vary the least, with the notion of “least” defined by the loss c . Thus, the representation characterizes a center by what it minimizes rather than by how it is computed.

The proof of Theorem 1 is constructive, and is available in Appendix A.1. Broadly speaking, Axioms 1 – 3 are regularity conditions shared with classical theories of means (Kolmogorov, 1930; Nagumo, 1930; Fishburn, 1986): Certainty Matching fixes the normalization $c(x, x) = 0$, Continuity delivers continuity of

⁹In the necessity direction of the proof, the first order condition holds for every admissible adjacent pair; so, the reduced dataset is valued y regardless of which pair is merged. Conversely, the sufficiency proof requires only one valid reduction per step.

¹⁰For instance, let $D = \{0, 3, 12; \frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}$ and take $\alpha = \frac{1}{3}$, so that $m(D) = \mu_{1/3}(D) = 3.6$. Since $x_1, x_2 < 3.6 < x_3$, the axiom prescribes low-side aggregation on the adjacent pair $(0, 3)$. Fixing any anchor $\bar{k} > 3.6$, say $\bar{k} = 12$, we solve for the weights $r_{3.6}(x)$ such that the $\frac{1}{3}$ -expectile of $\{x, \bar{k}; r_{3.6}(x), 1 - r_{3.6}(x)\}$ equals 3.6, obtaining $r_{3.6}(0) = \frac{7}{13} \approx 0.54$ and $r_{3.6}(3) = \frac{7}{8} = 0.88$. Applying Equation (2) yields $r_{3.6}(\tilde{x}) = \frac{2}{3}$, hence $\tilde{x} = 1.5$, and $D' = \{1.5, 12; \frac{2}{3}, \frac{1}{3}\}$.

c and existence of the minimizer, and Dominance guarantees strict convexity. Translation Invariance allows to shift the reference point y on the real line without any changes to the shape of c . The substance lies in the remaining two axioms: Odds Independence guarantees that the marginal loss recovered from binary comparisons is independent of the anchor against which it is measured; hence, it is what allows a *single* c — rather than a reference-dependent family of loss functions — to rationalize all binary datasets at once. Finally, Reduction occupies the role that associativity plays in the Kolmogorov—Nagumo characterization of quasi-linear means: it extends the binary representation to datasets of arbitrary size by requiring that collapsing a dataset one observation at a time leaves its center fixed.

Uniqueness of c up to a positive multiplicative constant ensures that the averaging behavior encoded in m recovers the loss that generates it. This is what permits to adopt the decision-theoretic approach proposed in Section 5: if c expresses a decision maker’s preferences over forecast errors, then observing how she averages numbers allows to recover such preferences.

Relative to the characterizations discussed in Section 2, the contribution of this paper lies in the direction of the argument: rather than fixing a functional and asking which scores are consistent for it, Theorem 1 places interpretable axioms on the location mapping and recovers the admissible class: the set of translation-invariant M -estimators (Huber, 1964).

Subsection 4.1 makes this concrete, showing how popular location indices — mean, median, mode, and their generalizations — relate to this representation.

4.1 Mean, median, mode.

A position index admits a variational representation exactly to the degree that its generating loss function belongs to $\mathcal{C}$. In this subsection, I show that the variational representation discussed above accommodates the standard location indices (mean, median, mode) to different degrees, and the pattern is instructive: expectiles (Newey and Powell, 1987),¹¹ — which include the mean — are generated by strictly convex losses, and lie inside the class. Quantiles — which include the median — are generated by losses that are convex

¹¹For any $\alpha \in (0, 1)$, the α -expectile of a distribution D is $\mu_\alpha(D) = \arg \min_\mu \mathbb{E}[\rho_\alpha(D - y)]$, with:

$$\rho_\alpha(u) = \begin{cases} \alpha u^2 & u \geq 0 \\ (1 - \alpha)u^2 & u < 0 \end{cases} \quad (3)$$

but not strictly so; hence, they sit on the boundary of $\mathcal{C}$, and are recovered only as a limit. On the other hand, the mode — generated by a discontinuous loss — falls outside the class altogether. So, the three cases help tracing the boundary of what the axioms can represent. This classification is formalized by the next result, where, for a given $D \in \mathcal{D}$, $\mu_\alpha(D)$ denotes its α -expectile, $q_\alpha(D)$ its α -quantile, and $M(D)$ its mode.

Proposition 1 (Expectiles, Quantiles, Mode.)

1. $\forall \alpha \in (0, 1), \exists c \in \mathcal{C}$ such that $m(D) = \mu_\alpha(D)$ for any $D \in \mathcal{D}$;
2. $\forall \alpha \in (0, 1), \exists \{c_m\}_{m \in \mathbb{N}} \subset \mathcal{C}$ such that (i) for any pair (x, y) , $\lim_{m \rightarrow \infty} c_m(x, y) = c(x, y)$ and (ii) $\arg \min_y \sum_{i=1}^n p_i c(x_i, y) = q_\alpha(D)$ for any $D \in \mathcal{D}$;
3. $\nexists c \in \mathcal{C}$ such that $m(D) = \mu_\delta(D) \forall D \in \mathcal{D}$.

The formal proof is available in Appendix A.2, and it consists in showing that the asymmetric squared losses generating expectiles belong to $\mathcal{C}$, the piecewise-linear functions generating quantiles do not but are the pointwise limit of a sequence that does, while the weighted 0–1 loss generating the mode is excluded from $\mathcal{C}$ altogether.

It is worth noting two features of expectiles and quantiles. First, for both families of indices, α usually generates an asymmetry between over- and under-prediction: unless $\alpha = 0.5$, the loss functions generating expectiles and quantiles penalize relatively more negative mistakes (i.e. $\alpha < 0.5$), or positive ones (i.e. $\alpha > 0.5$), for a given size.¹²

Second, while α fixes the direction in which errors are overweighted, the convexity of the loss determines how sharply the location index will react to outliers. This is where expectiles and quantiles differ: the expectile’s quadratic loss penalizes a mistake by its squared magnitude; so, an outlier will pull the index toward itself proportionally to its distance. On the other hand, the quantile’s linear loss only penalizes the side on which a residual falls; so, the index responds to the mass lying above and below it, but not to how far that mass is placed.

In words, expectiles are minimizers of asymmetric square loss functions, and they correspond to the sample mean for $\alpha = 0.5$. Intuitively, the α -expectile of D is the value that would be the mean if values above it occurred $\frac{\alpha}{1-\alpha}$ times as often as they actually do.

¹²Asymmetric losses have a sampling rationale: when data go missing at rates that differ by the sign of the error, the α -expectile and α -quantile of the observed distribution recovers the mean and the median of the latent one, with $\frac{\alpha}{1-\alpha}$ being the relative odds of positive versus negative mistakes (Philipps, 2022).

These two features, directionality and robustness, are nicely parametrized for expectiles and quantiles,¹³ but the underlying intuition is general. Any $c \in \mathcal{C}$ weights the two types of mistakes in some proportion, and penalizes their sizes at some rate. Because the primitive of this paper is the map m rather than the loss function, these features are inherited from the chosen index itself: by Theorem 1 the loss is recovered from m ; so, its asymmetry and convexity are implicitly defined by the averaging behavior of m . In this sense, the representation recovers some sort of *preferences over averages* — a disposition towards how errors of each sign and size should count when determining a location index. Section 5 builds on this intuition, interpreting the features of asymmetry and convexity of loss functions through a decision-theoretic lens.

V. ATTITUDES OVER MISTAKES

By functional translation invariance, $c(x, y) = c(0, y - x)$: the loss c depends on x and y only through the residual $e := y - x$. Note that $e > 0$ implies *over-prediction*, while $e < 0$ implies *under-prediction*. For the sake of clarity, let me define $\phi(e) := c(x, y)$.

Asymmetry. Then, I can define asymmetry of ϕ (of c) as follows:

Definition 2 (Local Asymmetry) *Fix a residual $e > 0$. The loss ϕ is:*

- *symmetric at e if $\phi(e) = \phi(-e)$,*
- *positively asymmetric at e if $\phi(e) > \phi(-e)$,*
- *negatively asymmetric at e if $\phi(e) < \phi(-e)$.*

It is worth emphasizing this is a local requirement: a loss may be positively asymmetric at one residual and negatively asymmetric at another.¹⁴ Global asymmetry is defined by requiring the conditions above to hold for all $e > 0$. I next proceed by proposing a quantifier for local asymmetry:

Definition 3 (Asymmetry Coefficient) *Fix a residual $e > 0$. The asymmetry coefficient of ϕ is:*

$$\mathcal{A}(e) = \log \left[\frac{\phi(e)}{\phi(-e)} \right], \quad \mathcal{A}_0 := \lim_{e \rightarrow 0^+} \mathcal{A}(e)$$

¹³They are measured by the asymmetry coefficient α and the power of the distance p , respectively.

¹⁴For instance, take $\phi(e) = e^2 + \frac{1}{4} \sin^3(x)$.

It is easy to see that $\mathcal{A}(e)$ is equal to zero whenever ϕ is symmetric at e , positive whenever ϕ is positively asymmetric at e , negative otherwise. In particular, $\mathcal{A}(e) \rightarrow +\infty$ denotes the limiting case of a loss function penalizing positive mistakes only (i.e. $\phi(e) \rightarrow +\infty$ or $\phi(-e) \rightarrow 0$), while $\mathcal{A}(e) \rightarrow -\infty$ implies a ϕ that penalizes negative mistakes only (i.e. $\phi(e) \rightarrow 0$ or $\phi(-e) \rightarrow +\infty$). Additionally, $\mathcal{A}_0$ — which I will refer to as *origin asymmetry* — measures asymmetry in the immediate neighborhood of a null mistake.¹⁵

I can provide a behavioral interpretation of the asymmetry of ϕ . To do so, let us suppose that D is a simple lottery, where each x_i denotes a monetary prize, and each p_i is its corresponding probability. Take a decision maker (henceforth, DM) who must set a *selling price* the lottery.¹⁶ When solving this task, we can imagine she tries to balance two possible sources of regret: setting a high price for a low realization (with the risk of not being able to sell the lottery), or setting a low price for a good one. If we are willing to interpret ϕ as a *regret function*, where regret is strictly increasing in the realized discrepancy $|x_i - y|$, then the DM solves exactly the minimization problem described in Theorem 1.¹⁷

Then, in this framework, $m(D)$ represents the selling price that minimizes expected regret,¹⁸ and the shape of ϕ is the primitive expression of the DM's preferences over different mistakes. In particular, the origin asymmetry $\mathcal{A}_0$ measures local risk attitudes. To see why, let me consider the DM's risk premium for lottery D :

$$\pi(D) = E(D) - m(D)$$

so that $\pi(D) > 0$ is risk aversion, $\pi(D) = 0$ is risk neutrality, and $\pi(D) < 0$ is risk seeking. To isolate the role of asymmetry, I evaluate $\pi(D)$ on a symmetric binary lottery $\bar{D} = \{-t, t; \frac{1}{2}, \frac{1}{2}\}$, so that $\pi(\bar{D}) = -m(\bar{D})$. Then — under mild regularity conditions — it is possible to show that:

$$\pi(D_t) = -\tanh\left(\frac{\mathcal{A}_0}{2}\right)t + o(t) \tag{4}$$

¹⁵For check (Koenker and Bassett Jr, 1978) and swoosh functions (Newey and Powell, 1987), it is easily showed that $\mathcal{A}(e)$ is constant for all e , — i.e. check and swoosh functions are globally (a)symmetric — and corresponds to the logit function of their asymmetry parameter α . This includes the origin asymmetry $\mathcal{A}_0$.

¹⁶The interpretation is theoretically the same if the DM is trying to *buy* the lottery, instead.

¹⁷For a more detailed discussion of this interpretation, check my other paper “When do certainty equivalents not measure risk preferences?”.

¹⁸This corresponds to the certainty equivalent of lottery D in the sense of Fishburn (1986).

Broadly speaking, Equation (4) maps the loss’ origin asymmetry to the sign of the risk premium.¹⁹ For instance, a loss with negative origin asymmetry — i.e. one that penalizes small downside discrepancies from the reference point more sharply than equal upside ones — generates a strictly positive premium: the DM sets a “conservative” certainty equivalent, and is risk averse. Positive origin asymmetry generates a negative premium and risk seeking. The intuition behind null and positive origin asymmetry is analogous.

Note that Equation (4) implies that the risk premium $\pi(D)$ is linear in t . In the taxonomy of Segal and Spivak (1997), this is first-order risk aversion: the risk premium is proportional to the standard deviation of the gamble rather than to its variance, and does not become negligible as the gamble shrinks.²⁰ Segal and Spivak (1997) show that first-order risk aversion requires a kink in the agent’s representation at the point of certainty. In this model, this can be found at $\phi(0)$.

Convexity. While asymmetry concerns with the distinction between over- and under-approximation, — that is, how ϕ responds to mistakes of different signs — a logically distinct feature concerns how it responds to mistakes of different magnitudes: in other words, how steeply the loss grows as the location index y moves further from observation x . This aspect is captured by the degree of convexity of ϕ . Unlike asymmetry, which entails a comparison between the two branches of the loss functions, convexity is a *within-branch* property. I measure convexity as follows:

Definition 4 (Convexity Coefficient) *Fix a residual $e \neq 0$. If ϕ is twice-differentiable at e , then the convexity coefficient of ϕ at e is:*

$$\kappa(e) := \frac{e \phi''(e)}{\phi'(e)}, \quad \kappa(0) := \lim_{e \rightarrow 0} \kappa(e)$$

Two aspects are worth pointing out. First, just like asymmetry, convexity is a local property, and usually varies across residuals.²¹ Second, even if the coefficient shares its functional form with the Arrow-Pratt coefficient of relative risk aversion, the resemblance is purely formal, and $\kappa(e)$ can’t be interpreted as

¹⁹For the swoosh loss, $-\tanh\left(\frac{A_0}{2}\right) = 2\alpha - 1$, so that $\alpha < 0.5$ implies risk aversion, $\alpha = 0$ implies risk neutrality, and $\alpha > 0.5$ implies risk seeking. $m(D)$ is the α -expectile of D .

²⁰This is a sharp difference from EU, which assumes second order risk aversion. As a result, a smooth EU agent is approximately risk neutral toward small risks.

²¹For the power family of loss functions, $\kappa(e)$ is constant, and is equal to $\theta - 1$.

a measure of second-order risk aversion. To see why, first note that the two coefficients are computed on different primitives over different arguments: relative risk aversion is the relative curvature of utility u a function of monetary prizes x ; and summarizes how the DM values money. On the other hand, the convexity coefficient is the relative curvature of loss ϕ as a function of the mistake e , and summarizes how the loss for mis-prediction grows with its size.

Second, convexity and Arrow-Pratt risk aversion act at different orders of small-gamble premia: while the former affects the first-order coefficient — i.e. the slope of the premium in the stake, and hence whether arbitrarily small gambles are rejected — the latter sets the second-order coefficient. Hence, convexity cannot be a measure of Arrow-Pratt risk aversion.

Rather, $\kappa(e)$ is a measure of *sensitivity to extremes*: it quantifies the degree to which ϕ penalizes larger residuals compared to smaller ones. In this context, extremes are defined in relative terms with respect to the location index y : to further away x is from y (in either direction), the more extreme x is. In other words, convexity determines the rate at which an observation's influence on the optimal value of y is amplified by its distance from y itself.

Indeed, the convexity coefficient is connected to the gross error sensitivity of m at D , the usual quantifier of robustness of estimators to outliers (Hampel, 1968).²² Since the influence function of m at D is proportional to the contribution evaluated at the contaminating residual, the gross-error sensitivity is finite if and only if the marginal loss is bounded or, equivalently, the tail convexity $\bar{\kappa} = \lim_{|e| \rightarrow \infty} \kappa(e)$ does not exceed zero. Thus, the coefficient traces a continuum of robustness: at $\kappa(e) = 0$, the contribution is bounded and m is robust (the quantile case); $\kappa(e) = 1$, the influence grows linearly, and m inherits the non-robustness of the mean; for $\kappa(e) > 1$, the influence grows super-linearly and m is anti-robust, i.e. its value is mainly driven by extreme observations.

²²The Gross Error Sensitivity (GES) of functional m at D is:

$$\gamma(m, D) = \sup_x |IF(x; m, D)|$$

where $IF(x; m, D) = \lim_{\epsilon \rightarrow 0} \frac{m[(1-\epsilon)D + \epsilon \delta_x] - m(D)}{\epsilon}$ is the Influence Function of m at D . In words, $\gamma(m, D)$ measures the highest influence that an infinitesimal contamination of the underlying distribution D with point x can have on the estimator m . While it is often infinite, signaling that m will account for any type of contaminations — including extreme ones, there exist some natural conditions under which the GES has a positive lower bound (Hampel, 1974). Hence, we can compare local robustness of estimators at by comparing their GES at D .

Identification. Conceptually, asymmetry and convexity are distinct features of ϕ : while the first shapes a directional attitude toward over- versus under-prediction, the second determines the rate at which the loss grows with the size of the mistake. Whether they can be successfully separated, however, is a different question. The answer is negative on two independent grounds: one structural, concerning the shape of ϕ itself, and one observational, concerning what a single forecast can reveal.

From a structural perspective, it must be noted that convexity is usually branch-dependent. I refer to this phenomenon as asymmetric convexity: in this case, the marginal role of each feature in determining the position of $m(D)$ is not defined.²³ Orthogonality is only recovered when ϕ is described by a unique convexity coefficient that is invariant to asymmetry.²⁴ As a result, the analyst cannot disentangle the marginal contribution of each feature to $m(D)$ from forecast data on binary datasets: since y acts as an endogenous reference point, observations from binary datasets will necessarily be split between one under-prediction and one over-prediction, making impossible to gather within-branch information.

The observational issue is even more fundamental, and holds regardless of whether ϕ displays asymmetric convexity or not. It is that sensitivity to extremes cannot be identified from the evaluation of a single dataset. The following result makes this precise:

Theorem 2 $\forall D \in \mathcal{D}$ and $\forall m(D) \in (x_1, x_n)$, $\exists \hat{\alpha} \in (0, 1)$ such that $m(D) = \mu_{\hat{\alpha}}(D)$.

The proof is available in Appendix A.3. In words, for any dataset D with a measure of centrality equal to $m(D)$, there exists an effective asymmetry level $\hat{\alpha} \in (0, 1)$ such that $m(D)$ coincides with $\hat{\alpha}$ -expectile of D .²⁵ Consequently, a robust index can be (mis)interpreted as a sensitive index with a biased directional weight.²⁶

²³An example is reported in Appendix B.

²⁴For instance, this is satisfied by swoosh functions.

²⁵It is important to notice that Theorem 2 should be read *per dataset*: it says that each $m(D)$ can be matched by *some* expectile, not necessarily the same one.

²⁶To see how, let the vector of annual incomes (in thousands of pounds) for five individuals be $D = \{20, 30, 40, 50, 460\}$. The mean is 120, the median is 40. However, the median also corresponds to the 6.52%–Expectile. Hence, evaluating D by its median is equivalent to assigning 14 times more weight to poorer individuals.

VI. CONCLUDING REMARKS

In this paper, I characterize location indices from an axiomatic perspective: rather than choosing a loss function and asking which location measure it elicits, I take the location mapping as the primitive of the model, and explore the conditions under which it can be represented as the minimizer of a continuous, strictly convex, translation-invariant expected loss. The representation I obtain is unique up to a positive multiplicative constant, and recovers the class of translation-invariant M -estimators.

The characterization also provides a behavioral interpretation of the underlying loss: since it is directly recovered from choice, its properties can be interpreted as revealed features of the averaging procedure. In particular, I show that its degree of origin asymmetry captures how over- and under-prediction are traded off, while convexity determines the extent to which extreme observations affect the reported measure of centrality. At the same time, the analysis shows that these features cannot be separately identified from a single evaluation.

More generally, the paper shows that location indices can be studied through the properties of averaging choices, without making any assumptions on the underlying loss *a priori*. Such perspective complements the existing literature on elicibility by providing an axiomatic route from observable averaging behavior to the loss function that rationalizes it.

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I. PROOFS

1.1 Theorem 1

Assume m admits a variational representation: $\exists c \in \mathcal{C}$ such that:

$$m(D) = \arg \min_y \sum_{i=1}^n p_i c(x_i, y) \quad (5)$$

for any $D \in \mathcal{D}$. To show Axiom 1, take any degenerate distribution $\{x; 1\}$. The axiom immediately follows from the fact that $c(x, y) \geq 0$ and $c(x, y) \Leftrightarrow x = y$.

To show Axiom 2, take any pair $D_k, D^* \in \ddot{\mathcal{D}}$ such that $D_k \rightarrow D^*$. Let $y_k = m(D_k)$ and $y^* = m(D^*)$. Note that strict convexity of $c(x, y)$ implies that $m(D)$ always exists and is unique; so, y_k and y^* are well defined. Also, $m(D) \in (x_1, x_n)$. As y_k belongs to a compact set, I can take any subsequence $I \subset \mathbb{N}$ and obtain that y_k for $k \in I$ has an accumulation point y' . By definition of y_k :

$$C(D_k, y_k) \leq C(D_k, y) \quad (6)$$

for any $y \in [x_1, x_n]$ and $k \in I$. As $c(x, y)$ is continuous, $C(D, y)$ is also continuous; so, Equation 6 in the limit becomes:

$$C(D^*, y') \leq C(D^*, y)$$

for any $y \in [x_1, x_n]$. As the minimizer is unique, this proves $y' = y^*$. In particular, this shows $y_k \rightarrow y^*$.

To prove Axiom 3, let $D_1 = \{x_1, x_2; p, 1 - p\}$ and $D_2 = \{z_1, z_2; q, 1 - q\}$ such that $D_1 \succ_{FOSD} D_2$. Then, $\exists \hat{D}_1, \hat{D}_2 \in \ddot{\mathcal{D}}$ defined on the same probability space such that $\hat{D}_1 =_{FOSD} D_1$, $\hat{D}_2 =_{FOSD} D_2$ and $P(\hat{D}_1 > \hat{D}_2) = 1$ (Theorem 1.A.1, Shaked and Shanthikumar, 2007). In other words, there exist a $k \in \mathbb{N}$, with $k \geq \max\{m, n\}$, and a vector $\delta = (\delta_1, \delta_2)$ with $\delta_i \geq 0$ for each i (at least one strict to avoid triviality) such that:

$$\hat{D}_1 = \{w_1 + \delta_1, w_2 + \delta_2; r, 1 - r\}$$

$$\hat{D}_2 = \{w_1, w_2; r, 1 - r\}$$

It is easy to see that $\hat{y}_1^* = y_1^*$ and $\hat{y}_2^* = y_2^*$. Next, note that $\hat{y}_2^*$ must satisfy:

$$\sum_{i=1}^k r_i c'(w_i, \hat{y}_2^*) = 0 \quad (7)$$

It follows that $\sum_{i=1}^k r_i c'(w_i + \delta_i, \hat{y}_2^*) < 0$, which implies $\hat{y}_1^* > \hat{y}_2^* \Rightarrow y_1^* > y_2^*$.

Axiom 4 follows from $c(x, y) = c(x + k, y + k)$ for any k . To see why, take any $D \in \ddot{\mathcal{D}}$ and let $m(D) = y$.

Take also any other $k, y' \in \mathbb{R}$. Then, I can write:

$$\begin{aligned} C(D, y) &\leq C(D, y') \\ \sum_{i=1}^n p_i c(x_i, y) &\leq \sum_{i=1}^n p_i c(x_i, y') \\ \sum_{i=1}^n p_i c(x_i + k, y + k) &\leq \sum_{i=1}^n p_i c(x_i + k, y' + k) \\ C(D + k, y + k) &\leq C(D + k, y'') \end{aligned}$$

for all $y'' \in \mathbb{R}$. Hence, $m(D + k) = m(D) + k$.

Axiom 5 is easily shown by observing that $m(D) = m(D') = m(D'') = m(D''') = y$ implies that y must satisfy the following system of equations:

$$\begin{cases} p c'(\underline{x}, y) + (1 - p) c'(\bar{x}, y) = 0 \\ q c'(\underline{x}, y) + (1 - q) c'(\bar{\bar{x}}, y) = 0 \\ r c'(\underline{\underline{x}}, y) + (1 - r) c'(\bar{x}, y) = 0 \\ s c'(\underline{\underline{x}}, y) + (1 - s) c'(\bar{\bar{x}}, y) = 0 \end{cases}$$

which is only satisfied whenever $\frac{q}{1-q} \cdot \frac{1-p}{p} = \frac{s}{1-s} \cdot \frac{1-r}{r}$.

To show Axiom 6, first assume $y = x_k$ for some $x_k \in \text{supp}(D)$. The first order condition for D , together with the fact that $c'(x_k, x_k) = 0$, imply that $\sum_{i \neq k} \frac{p_i}{1-p_k} c'(x_i, y) = 0$. Hence, $D' = \left\{ \{x_i\}_{i \neq k}; \left\{ \frac{p_i}{1-p_k} \right\}_{i \neq k} \right\}$ is also valued y .

Next, consider the case of $y \in (x_k, x_{k+1})$ for some $x_k, x_{k+1} \in \text{supp}(D)$, and let D' be a dataset obtained by high-side aggregation. As each item admits a variational representation, the following set of equations must be satisfied:

$$\left\{ \begin{array}{l} [1 - q_y(x_i)]c'(k, y) + q_y(x_i)c'(x_i, y) = 0 \\ [1 - q_y(\tilde{x})]c'(k, y) + q_y(\tilde{x})c'(\tilde{x}, y) = 0 \\ [1 - q_y(x_{i+1})]c'(k, y) + q_y(x_{i+1})c'(x_{i+1}, y) = 0 \\ p_1c'(x_1, y) + \dots + p_ic'(x_i, y) + p_{i+1}c'(x_{i+1}, y) + \dots = 0 \\ p_1c'(x_1, y) + \dots + (p_i + p_{i+1})c'(\tilde{x}, y) + \dots = 0 \end{array} \right. \quad (8)$$

It is easy to show that System (8) implies Equation 2 for any k . The proof for low-side aggregation is analogous.

For sufficiency, I adopt a constructive approach: I show that I can construct a (marginal) loss function $c \in \mathcal{C}$ that rationalizes the map m . I start by showing the result for $n = 2$, and then extend it to non-binary datasets. Throughout the proof, I denote any value $x < 0$ as x^- , and any value $x > 0$ as x^+ .

First, let me normalize $c'(1, 0) = -1$. For each x^- , let $D^- = \{x^-, 1, p^-\}$ be the dataset such that $m(D^-) = 0$.

The existence and uniqueness of p^- are guaranteed by Axioms 1-3, as showed by Lemma 1.

Lemma 1 *Take any $x^-, x^+, y \in \mathbb{R}$ such that $x^- < y < x^+$. If m satisfies Axioms 1-3, then $\exists ! p \in (0, 1)$ such that $S(\{x^-, x^+, p\}) = y$.*

Proof. Let $f : [0, 1] \rightarrow \mathbb{R}$ be such that $f(p) = m(\{x^-, x^+; p\})$, where x^-, x^+ are fixed. Axiom 2 ensures that $f(p)$ is continuous. Also, observe that Axiom 1 implies that $f(0) = x^+$ and $f(1) = x^-$. Then, $\exists \hat{p} \in (0, 1)$ such that $m(\{x^-, x^+, \hat{p}\}) = y$ by Bolzano's theorem.

To prove uniqueness, assume there exists two values, $\hat{p}_1$ and $\hat{p}_2$, such that the condition above is satisfied. This means that $m(\{x^-, x^+ : \hat{p}_1\}) = m(\{x^-, x^+ : \hat{p}_2\})$. But this violates Axiom 3. ■

Using the first order condition for D^- , I obtain:

$$c'_l(x, 0) = \frac{1 - p^-}{p^-} \quad (9)$$

In particular, note that $\lim_{x^- \rightarrow 0^-} c'_l(x, 0) = 0$. Also, $c'_l(x, 0)$ is unique up to a positive multiplicative

constant.²⁷

Similarly, fix a value $\hat{x}^-$. For each $x^+ > 0$, let $D^+ = \{\hat{x}^-, x^+, p^+\}$ be an item such that $m(D^+) = 0$. The first order condition, together with Equation 9, implies:

$$c'_r(x, 0) = -\frac{p^+}{1-p^+} \cdot \frac{1-\hat{p}^-}{\hat{p}^-} \quad (10)$$

where $\hat{p}^-$ is the value such that $\{\hat{x}^-, 1; \hat{p}^-\}$ is mapped to 0. Again, observe that $\lim_{x^+ \rightarrow 0^+} c'_r(x, 0) = 0$, and that $c'_r(x, 0) = 0$ is unique up to a positive multiplicative constant.

Importantly, the choice of $\hat{x}^-$ to derive Equation (10) is irrelevant by Axiom 5. To see why, take any $\hat{x}^+ > 1$ and $\tilde{x}^- < \hat{x}^-$, and define $D = \{\hat{x}^-, 1; \hat{p}^-\}$, $D' = \{\hat{x}^-, x^+, \hat{p}^+\}$, $D'' = \{\tilde{x}^-, 1; \tilde{p}^-\}$ and $D''' = \{\tilde{x}^-, \hat{x}^+; \tilde{p}^+\}$ such that $m(D) = m(D') = m(D'') = m(D''') = 0$. The procedure described above applied to D and D' delivers:

$$c'(x, 0) = \begin{cases} \frac{1-p^-}{p^-} & \text{if } x < 0, \\ 0 & \text{if } x = 0, \\ -\frac{\hat{p}^+}{1-\hat{p}^+} \cdot \frac{1-\hat{p}^-}{\hat{p}^-} & \text{if } x > 0 \end{cases}$$

The same procedure applied to D'' and D''' delivers:

$$c'(x, 0) = \begin{cases} \frac{1-p^-}{p^-} & \text{if } x < 0, \\ 0 & \text{if } x = 0, \\ -\frac{\tilde{p}^+}{1-\tilde{p}^+} \cdot \frac{1-\tilde{p}^-}{\tilde{p}^-} & \text{if } x > 0 \end{cases}$$

²⁷To see why, take any $x > 0$ different from 1, and let it be denoted by x^+ . Let $D'^- = \{x^-, x^+, p'^-\}$ be the item such that $m(D'^-) = 0$. Finally, let $k > 0$ be such that $c'(x^+, 0) = -k$. Then, it must be that $\frac{1-p^-}{p^-} = k \cdot \frac{1-p'^-}{p'^-}$.

It is easy to see these two expressions are the same because of Axiom 5. Hence, I can define:

$$c'(x, 0) = \begin{cases} c'_l(x, 0) & \text{if } x < 0, \\ 0 & \text{if } x = 0, \\ c'_r(x, 0) & \text{if } x > 0 \end{cases} \quad (11)$$

Finally, I use Axiom 4 to move the second argument of c' from 0 to a generic value y . To reach this goal, note that Axiom 4 guarantees that $c'(x, 0) = c'(x + y, y)$ for any y . With a slight abuse of notation, I now redefine $x = x + y$, and define the generic marginal cost function as follows:

$$c'(x, y) = \begin{cases} c'_l(x, y) & \text{if } x < y, \\ 0 & \text{if } x = y, \\ c'_r(x, y) & \text{if } x > y \end{cases} \quad (12)$$

Next, I show that c' as defined in Equation (12) is continuous in each argument, and strictly increasing in its second argument. To perform this aim, let $f : \mathbb{R}^3 \rightarrow [0, 1]$ be a map that, to each triple (x^-, y, x^+) where $x^- \leq y \leq x^+$, assigns the value $p \in [0, 1]$ such that $m(\{x^-, x^+; p\}) = y$. It is easy to see that $f(x^-, 0, 1) = p^-$ and $f(\hat{x}^-, x^+, 1) = p^+$. Lemma 2 shows that, under Axioms 2 and 3, f is continuous in each variable.

Lemma 2 *If m satisfies Axioms 2 and 3, then $f(x^-, y, x^+) \rightarrow p$ is continuous in each variable.*

Proof. Let $m(p) = m(\{x^-, x^+; p\})$ for a fixed pair (x^-, x^+) . Axioms 2 and 3 ensure that $m(p)$ is continuous and strictly decreasing; hence, $m(p)$ has a continuous inverse: $\exists g : [x^-, x^+] \rightarrow [0, 1]$ such that $g(y) = p$ and g is continuous. But $g(y) = f(x^-, y, x^+)$ for the fixed pair (x^-, x^+) ; so, f is continuous in y . The other two cases are analogously showed. ■

Lemma 2, together with the limiting behavior of c' when $x^- \rightarrow y \rightarrow x^+$, ensures that c' is continuous in each argument.

To see why c' is strictly increasing in its second argument, first observe that $c'(x, y)$ is strictly decreasing in its first argument. This is a direct consequence of Lemma 3.²⁸

Lemma 3 *If m satisfies Axiom 3, then $f(x^-, y, x^+)$ is strictly increasing in both x^- and x^+ .*

Proof. Let $p(x^-) = f(x^-, y, x^+)$ for a fixed pair (y, x^+) , and take two negative values $x_1^- > x_2^-$. Assume by contradiction that $p(x_1^-) \leq p(x_2^-)$. This implies $\{x_1^-, x^+; p(x_1^+)\} \succ_{FOSD} \{x_2^-, x^+; p(x_2^+)\}$, violating Axiom 3. The other case is analogously showed. ■

Then, take three values $x_1, x_2, y_1 \in \mathbb{R}$ with $x_1 > x_2$, and define $\epsilon \equiv x_1 - x_2 > 0$. Also, let $y_2 = y_1 + \epsilon$. As c' is strictly decreasing in its first argument, it holds that:

$$c'(x_1, y_1) < c'(x_2, y_1)$$

$$c'(x_1, y_1) < c'(x_1 - \epsilon, y_1)$$

$$c'(x_1, y_1) < c'(x_1, y_1 + \epsilon)$$

$$c'(x_1, y_1) < c'(x_1, y_2)$$

where the passage from the second to the third step is enabled by Axiom 4. But this proves that c' is strictly increasing in its second argument.

As $c'(x, y)$ is continuous in its second argument, it is integrable. Hence, I consider:

$$c(x, y) = \int c'(x, y) dy \tag{13}$$

Observe that Axiom 1 requires the constant of integration to be equal to zero.²⁹

I now show that $c(x, y) \in \mathcal{C}$. First, fix x . As $c'(x, y)$ is negative if $y < x$, null if $y = x$, positive otherwise, $c(x, y)$ is strictly decreasing if $y < x$, zero if $y = x$, strictly increasing otherwise. It follows that $c(x, y) \geq 0$, with equality if and only if $y = x$.

²⁸Simply apply the definition of strictly increasing function.

²⁹By Axiom 1, $c(x, x) = \int dy + k = 0$. This implies $k = 0$.

Also, c satisfies functional translation invariance by construction, as showed below:

$$\begin{aligned} c(x, y) &= \int c'(x, y) dy \\ c(x, y) &= \int c'(x + k, y + k) dy \\ c(x, y) &= c(x + k, y + k) \end{aligned}$$

Finally, since c' is continuous in each variable, c is also continuous in each variable. Similarly, as c' is strictly increasing, c is strictly convex (Rockafellar, 1997; Theorem 24.1).

Next, I extend the result to non-binary items inductively. Let $D = \{x_1, x_2, x_3; p_1, p_2, 1 - p_1 - p_2\}$, with $m(D) = y$. If $y = x_2$, then $D' = \{x_1, x_3; \frac{r}{1-s}, \frac{1-r-s}{1-s}\}$ is also valued x_2 by Axiom 6. Since D' is binary, it is represented by the function constructed above; hence, the following condition holds:

$$\frac{r}{1-s} c'(x_1, y) + \frac{1-r-s}{1-s} c'(x_3, y) = 0$$

This equation, together with the fact that $c'(x_2, y) = 0$, implies that c also rationalizes the map m for a ternary item whenever $y = x_2$.

If $y \in (x_2, x_3)$, then $D' = \{\tilde{x}, x_3; r + s\}$ is also valued y by Axiom 6. As $x_1 < \tilde{x} < x_2$ and c' is strictly decreasing in its first argument, $\exists \alpha \in (0, 1)$ such that $c'(\tilde{x}, y) = \alpha c'(x_1, y) + (1 - \alpha) c'(x_2, y)$. This, together with Equation 12, implies $\alpha = \frac{r}{r+s}$ (Axiom 6). Then, the first order condition of D' leads to:

$$r c'(x_1, y) + s c'(x_2, y) + (1 - r - s) c'(x_3, y) = 0$$

So, c also rationalizes the map m for a ternary item whenever $y \in (x_2, x_3)$. The proof for $y \in (x_1, x_2)$ is analogous.

Finally, assume the result holds (i.e. the cost function from Equation (12) rationalizes the map m) for any item of size $(n - 1)$. By following the same steps at above it is possible to show that, under this condition, c rationalizes the map for items of size n as well.

1.2 Proposition 1

By definition, the α -expectile of a distribution is the minimizer of a cost function defined as in Equation 1 with c corresponding to the so-called “swoosh” function (Newey and Powell, 1987):

$$c_\alpha(x, y) = \begin{cases} (1 - \alpha)(y - x)^2 & \text{if } x \leq y, \\ \alpha(x - y)^2 & \text{if } x > y \end{cases}$$

where $\alpha \in (0, 1)$. It is easy to check that $c_\alpha(x, y) \in \mathcal{C}$. Hence, the result holds.

The α -quantile of a distribution is the minimizer of a cost function defined as in Equation (1) with c corresponding to the so-called “check” function (Koenker and Bassett Jr, 1978). Take the following function:

$$c_m(x, y) = \begin{cases} (1 - \alpha)(y - x)^{(1 + \frac{1}{m})} & \text{if } x \leq y, \\ \alpha(x - y)^{(1 + \frac{1}{m})} & \text{if } x > y \end{cases}$$

where $\alpha \in (0, 1)$ and $m \in \mathbb{N}$. It is easy to check that $c_m(x, y) \in \mathcal{C}$. As $m \rightarrow \infty$, $c_m(x, y)$ converges exactly to the “check” function. Hence, the result holds.

The mode of a distribution is the minimizer of a cost function defined as in Equation (1) with c corresponding to the Dirac measure: $c_\delta(x, y) = \mathbf{1}_{\{x\}}(y)$. In words, c_δ is equal to 1 if $x = y$, to 0 otherwise. Being discontinuous, $c_\delta \notin \mathcal{C}$. Hence, the result holds.

1.3 Theorem 2

Let $\mu_\alpha(D) \in [x_1, x_n]$ denote the α -expectile of D . Being the optimal predictor under an asymmetric least squares loss function, $\mu_\alpha(D)$ must satisfy:

$$\alpha \sum_{x_i > \mu_\alpha(D)} p_i[x_i - \mu_\alpha(D)] = (1 - \alpha) \sum_{x_i \leq \mu_\alpha(D)} p_i[\mu_\alpha(D) - x_i] \quad (14)$$

Take any $m \in [x_1, x_n]$, and let $A(m) := \sum_{x_i > m} p_i(x_i - m)$ and $B(m) := \sum_{x_i \leq m} p_i(m - x_i)$. By letting $\mu_\alpha(D) = m$, Equation (14) implies:

$$\alpha(m) = \frac{B(m)}{A(m) + B(m)} \in [0, 1]$$

Fix an index $k \in \{1, \dots, n-1\}$ and consider $m \in (x_k, x_{k+1})$. Set

$$P_L := \sum_{i \leq k} p_i, \quad P_R := \sum_{i > k} p_i, \quad D := \sum_{i \leq k} p_i x_i, \quad C := \sum_{i > k} p_i x_i.$$

For such m no atom equals m , so the sums defining $A(m)$ and $B(m)$ simplify to

$$B(m) = \sum_{i \leq k} p_i (m - x_i) = mP_L - D, \quad A(m) = \sum_{i > k} p_i (x_i - m) = C - mP_R.$$

Hence on (x_k, x_{k+1}) we have the rational-linear form

$$\alpha(m) = \frac{B(m)}{A(m) + B(m)} = \frac{mP_L - D}{(C - D) + m(P_L - P_R)}.$$

Differentiate with respect to m . Using $A'(m) = -P_R$ and $B'(m) = P_L$ gives

$$\alpha'(m) = \frac{B'(A+B) - B(A'+B')}{(A+B)^2} = \frac{P_L(A+B) - B(P_L - P_R)}{(A+B)^2} = \frac{P_L A + B P_R}{(A+B)^2}.$$

The numerator equals the strictly positive double sum

$$P_L A + B P_R = \sum_{i \leq k} \sum_{j > k} p_i p_j (x_j - x_i) > 0,$$

since each $p_i p_j > 0$ and $x_j - x_i > 0$ for $j > i$. Thus $\alpha'(m) > 0$ on every open gap (x_k, x_{k+1}) .

Finally, note that for each fixed i , the functions $m \mapsto (x_i - m)$ and $m \mapsto (m - x_i)$ are continuous. Since $A(m)$ and $B(m)$ are finite weighted sums of such terms with positive weights p_i , both $A(m)$ and $B(m)$ are continuous functions on $\mathbb{R}$. Because the denominator $A(m) + B(m) = \sum_i p_i |x_i - m|$ is continuous and strictly positive for any nondegenerate distribution (i.e. not all mass concentrated at a single point), the quotient

$$\alpha(m) = \frac{B(m)}{A(m) + B(m)}$$

is also continuous on $\mathbb{R}$, and in particular on $[x_1, x_n]$.

At each atom x_k , $A(m)$ and $B(m)$ remain continuous because the contribution from the term where $x_i = m$ vanishes (it multiplies a zero difference). Therefore $\alpha(m)$ has no jump at any x_k ; the left and right limits coincide.

Since we already showed that $\alpha'(m) > 0$ for every m strictly between consecutive atoms, $\alpha(m)$ is strictly increasing on each open interval (x_k, x_{k+1}) . By continuity, the values of α at the boundaries connect these intervals smoothly, ruling out any downward jump. Hence $\alpha(m)$ is strictly increasing on the entire domain $[x_1, x_n]$.

Finally, observe that:

$$\alpha(x_1) = \frac{0}{\sum_{i>1} p_i(x_i - x_1)} = 0, \quad \alpha(x_n) = \frac{\sum_{i<n} p_i(x_n - x_i)}{\sum_{i<n} p_i(x_n - x_i)} = 1,$$

Combined with the fact that $\alpha(m)$ is continuous and strictly increasing on $[x_1, x_n]$, it follows that $\alpha : [x_1, x_n] \rightarrow [0, 1]$ is bijective; equivalently, for every $m \in [x_1, x_n]$ there exists a unique $\alpha \in [0, 1]$ (namely $\alpha = \alpha(m)$) such that $\mu_\alpha(D) = m$.

II. ASYMMETRIC CONVEXITY

Let ϕ be defined as follows:

$$\phi_a(e) = \begin{cases} |e|^\theta & \text{if } e < 0, \\ |e|^{\theta+a} & \text{if } e \geq 0 \end{cases}$$

for some $\theta > 1$ and $a \geq 0$. In this case, a raises the curvature of the upside branch of ϕ relative to the downside, leaving the downside fixed. If $a = 0$, then the two branches coincide, and $\phi_0(e) = |e|^\theta$ becomes the symmetric power loss. I first show that $\phi \in \mathcal{C}$:

- *Normalization*: both branches vanish as $e \rightarrow 0$; so, $\phi_a(0) = 0$. ϕ_a is continuous at 0 and trivially continuous elsewhere; hence, it is continuous. Also, $\phi_a > 0$ for $e \neq 0$.
- *Functional translation invariance*: satisfied by construction.

- *Strict convexity*: the first order derivative of ϕ_a is:

$$\phi'_a(e) = \begin{cases} -\theta|e|^{\theta-1} & \text{if } e < 0, \\ (\theta + a)|e|^{\theta+a-1} & \text{if } e \geq 0 \end{cases}$$

with $\phi'_a(0^-) = \phi'_a(0^+) = 0$ since both $\theta > 1$ and $\theta + a > 1$. Thus, ϕ'_a is continuous, equals 0 only at $e = 0$, and is strictly increasing in e . It follows that ϕ_a is strictly convex.

Next, I compute the relative convexity and asymmetry coefficients for each branch. I obtain that:

$$\kappa^-(e) = \theta - 1 \quad \kappa^+(e) = \theta + a - 1 \quad \mathcal{A}(e) = a \ln e$$

where $\kappa^-(e) \equiv \kappa(e|e < 0)$ and $\kappa^+(e) \equiv \kappa(e|e \geq 0)$. Note that both convexity coefficients are constant in e , but they differ across branches. So, ϕ exhibits asymmetric convexity for every $a > 0$: the curvature is determined jointly by the interaction of two different parameters, θ and a . Moreover, both the asymmetry $\mathcal{A}(e)$ and the difference in branches' curvature $\kappa^+(e) - \kappa^-(e)$ are governed by the same parameter a : increasing a increases both. Therefore, convexity at the point e is not invariant to asymmetry: we cannot increase the asymmetry of ϕ_a without increasing its upside curvature.

It must be stressed that this behavior is not a feature of asymmetry per se. To see why, consider this alternative loss function:

$$\psi_a(e) = \begin{cases} r_1|e|^\theta & \text{if } e < 0, \\ r_2|e|^\theta & \text{if } e \geq 0 \end{cases}$$

Under this formulation, the loss function ψ_a is still asymmetric: $\mathcal{A}(e) = \frac{r_2}{r_1}$ for all e . However, the asymmetry parameters r_1 and r_2 cancel out when computing convexity coefficients. Asymmetric convexity is precisely the departure from this multiplicative structure: it introduces asymmetry in the curvature rather than the scale of the loss.